



MarketLab

Every historical result has inspectable inputs and accounting.

MarketLab combines portfolio planning with saved historical experiments. A person can aggregate manually entered accounts and debts, validate stock-price history, compare a next-close trend rule with buying and holding, inspect costs and assumptions, and save an experiment that can be independently replayed.

The problem

A chart is not enough to trust a backtest. Imported prices, cash arithmetic, timing assumptions and saved inputs need to remain inspectable. In a background service, a crashed or stale worker must not overwrite the authoritative result.

What you can do

Planning before complexity A fictional first-run example shows how trading costs change the same investment. Private workspaces support goals, fee projections and contribution or stress scenarios.	Historical stock pipelines Server Alpha Vantage, browser Alpha Vantage and CSV imports converge on validation and immutable owner-scoped snapshots with explicit provenance.
Saved, exact backtests BigInt money and share arithmetic, declared events and causal next-close execution preserve the inputs behind each report.	Verification and reviewed sharing An owner may request C++ checks or independent Python replay. Separate reviewed sharing publishes an allowlisted, expiring, revocable summary.

How it works

01 / Import Fetch daily closes on demand or import one symbol's historical CSV with source, currency and adjustment declarations.	02 / Validate and freeze Reject malformed observations; save a provenance-bound, owner-scoped D1 snapshot with a SHA-256 identity.
03 / Calculate and save Bind the snapshot and event records to exact accounting and next-close execution; freeze the settings, inputs and result fingerprint.	04 / Verify or share Separate owner-triggered branches submit fenced Node/MongoDB jobs, request Python replay or publish a reviewed redacted summary.

TECHNOLOGY / TypeScript · React · Cloudflare Workers · Cloudflare D1 · Node.js · MongoDB · C++ · Python

Decisions that shape the system

Exact money, explicit execution

Cash uses BigInt cents and holdings use millionths of shares. Signals execute at the next supplied close, with defined fees, slippage, splits and dividends.

Separate runtime responsibilities

The hosted React/TypeScript application stores authoritative workspaces and reports in Workers/D1. A separate Node/MongoDB service owns copied research inputs and background jobs.

Fence stale workers

MongoDB-clock leases and tokens govern heartbeats and finalization. A crashed calculation can be retried without allowing an older worker to replace the current durable result.

Observed validation

Three local service-workload runs completed 3,000 verified jobs with zero failures, combined throughput 39.96 jobs/s and client-observed p99 319.77 ms. The workload used two calculation workers, eight closed-loop clients and 500 observations per instrument.

Scope of this evidence

These are local workload measurements, not production capacity or live-site latency. Data imports are user-triggered historical paths, not a streaming feed. Browser-imported provider data retains distinct provenance. Backtests are research comparisons, not forecasts.

Explore the implementation

Open MarketLab

marketlab-portfolio.michaelbaffour240306.chatgpt.site

Try the fictional report

marketlab-portfolio.michaelbaffour240306.chatgpt.site/example

Source code

github.com/michaelbawuah/MarketLab

Stock pipeline diagram

github.com/michaelbawuah/MarketLab/blob/main/docs/media/stock-data-pipeline.svg

Independent replay design

github.com/michaelbawuah/MarketLab/blob/main/docs/independent-replay.md

Service measurement methodology

github.com/michaelbawuah/MarketLab/blob/main/docs/service-performance.md

This brief explains the project using its public source and retained validation records. It is an engineering overview, not a research publication.